

Maintenance Manual: Alpha-X Industrial Pump

Document Version: 2.1 | Date: October 2025

1. Overview

The Alpha-X Industrial Pump is a heavy-duty centrifugal pump designed for high-pressure fluid transfer in manufacturing environments. It operates at a standard 3600 RPM and requires consistent thermal management.

2. Routine Maintenance Schedule

- Daily: Check for fluid leaks around the primary seal. Monitor vibration levels (should not exceed 4.5 mm/s).
- Weekly: Inspect the drive belt tension. Check motor casing temperature (normal operating range: 45C - 65C).
- Monthly: Lubricate the main shaft bearings using Grade 2 lithium grease. Test the emergency shut-off valve.
- Annually: Perform full impeller inspection and replace O-rings.

3. Troubleshooting Guide

High Vibration Levels (>5.0 mm/s)

High vibration is typically caused by bearing wear or impeller imbalance.

Action: Immediately reduce RPM to 1800. Inspect the mounting bolts for looseness. If bolts are tight, schedule a bearing replacement.

Temperature Spikes (>75C)

If the motor casing exceeds 75C, the cooling fins may be obstructed, or the bearing lacks lubrication.

Action: Shut down the pump. Clean the cooling fins with compressed air. Apply 15g of lithium grease

to the bearing housing.

4. Error Codes

- ERR-01 (Thermal Overload): Motor winding temperature is critical. Wait 30 minutes before restarting.
- ERR-02 (Flow Blockage): Intake pressure is below 1.2 bar. Check suction filters.